

NATIONAL CHILDREN'S SCIENCE CONGRESS

Theme: Science and Innovation for Sustainability

Sub-themes: Waste Management | Food, Agriculture & Health

Project Report - Form-A

AGRINOVA

Turning crop residue into income - not smoke, not drain waste

Predict. Protect. Prosper. Recycle.

Group members	Shikha Sharma . Samriddhi Ghosh
Class	XI (upper age group)
School	Kendriya Vidyalaya Dum Dum, Kolkata
Language	English
Survey area	Madhyamgram & Barasat wards . Punjab comparison belt
Sample size	n = 100 farm households (representative stratified sample)
Live desk	https://agrinova-hazel.vercel.app
Guide teacher	_____

Abstract

AgriNova asks why rice straw still leaves the farm as smoke in the north-west and as mixed drain waste in peri-urban West Bengal when compost pads, paper mills and biomass gates already buy the same material. The missing step is a shared listing - quantity, moisture, neighbourhood - that a smallholder and a nearby gate can both see.

This Form-A study combines published CRM / SWM context with a representative field survey of n = 100 farm households: 35 in Madhyamgram localities, 35 in Barasat wards, and 30 in the Punjab phone belt, stratified by landholding band (economic proxy). Plant-floor and ward-level conversations guided the design of AgriNova (<https://agrinova-hazel.vercel.app>), demonstrated on the same clusters.

Key result: Burned 24%, stacked/dumped 40%, mixed waste 20%, sold 16%. 81 of 100 would list if pickup came in five days (~650.4 t straw ~ Rs 455,280). The gap is not another scheme pamphlet - it is moisture-aware matching and a visible trolley date. Carbon figures use 1.5 tCO₂e/t as a working estimate, not certified credits.

Name & address of Guide Teacher: _____

Why this project?

Burned or dumped straw is wasted biomass, dirty air, and zero farm cash. Two clocks fail at one point: the farmer has no plant number and the plant has no small lot. From Kendriya Vidyalaya Dum Dum we can walk Madhyamgram-Barasat wards and hear the burn belt by phone. AgriNova is the listing this survey tests at meaningful sample size.

PART I - Selection of the problem and basic information

1. Selection of the problem

The focal theme asks for science that makes a local waste stream less harmful. The stream we can reach is peri-urban paddy leftover in Madhyamgram and Barasat (North 24 Parganas): wet stacks, drain push, and mixed municipal biodegradable waste. The Punjab belt enters as a comparison stratum because that is the published burn clock - and one listing must serve both.

We selected this problem because (i) the material is the same - rice straw; (ii) Bengal wards fail by dump/mix while the phone belt fails by burn; (iii) a municipal compost pad already exists while drains still take mixed straw; (iv) a Class XI team can close a representative sample of 100 households across wards and landholding bands, then demonstrate AgriNova.

1.1 Hypothesis

If a smallholder can list last season's straw with a moisture note and is promised pickup inside five days, then selling (or gifting to a plant) will replace burning or dumping as the default, provided the net gate price after freight is positive.

1.2 Objectives

1. Record residue fate across a representative sample of 100 farm households.
2. Identify the main barrier between field and plant gate, by zone and landholding band.
3. Test whether plant-floor staff will lift small lots when moisture is declared.
4. Document what ward / sanitary registers currently capture.
5. Field-test AgriNova on the same clusters and propose an abatement system.

1.3 Need and relevance

Madhyamgram and Barasat sit inside the Kolkata metropolitan stream: SWM Rules, 2016 already ask ULBs to process biodegradable waste, yet farm leftover rarely becomes a specified lot. Punjab's short sowing window is the comparison clock. The need is one Bangla/Punjabi listing desk that both clocks can use - without putting farmer phones on a municipal screen.

2. Basic information about the element chosen

The element is rice straw (paddy residue), sometimes with wheat stubble.

- Quantity: published residue-to-grain band ~ 1.8-2.4 t/acre; working figure used = 2.0 t/acre.
- Use: lignocellulose for biomass, paper furnish, or clean compost feedstock.
- Burning: near-zero farm cash; particulate load on the winter airshed (literature context).
- Dumping / mixing: wrong specification for SWM compost pads; becomes a conservancy complaint.

Element	Why it matters	How recorded
Residue fate	Burn / dump / sell / mix	One choice
Harvest-sow window	Clock that makes burning rational	Four bands
Own buyer phone	Matching gap	Yes / broker / none
Main barrier	Cause, not slogan	One choice
5-day pickup	Product test	Yes / maybe / no
Landholding band	Economic strata for sampling	S / M / L acres
Register contents	What ULB already sees	Officer note

Table 1. Data elements fixed before fieldwork.

3. Previous studies (defining the problem)

NPMCR (2014) and CRM machinery notes map surplus and in-situ options. SWM Rules, 2016 map urban biodegradable processing. SAFAR / IITM and CPCB episode notes give the public PM2.5 context for peak burn days. MSP cards explain why grain is sold and straw is treated as a nuisance; biomass tenders often sit near Rs 600-1200/t. None of these create a place where a 2-6 acre household, a hungry compost pad, and a ward councillor share one lot. That absence is what the survey measures.

4. Analysis of the problem and measurement

1. Farmer clock - short dry window (Punjab belt) vs longer wet stall (Bengal wards).
2. Plant clock - large tonnage at a moisture cap; small lots are invisible without aggregation.
3. Officer clock - complaints and mixed-waste lift, not tonnes utilised as a specified lot.

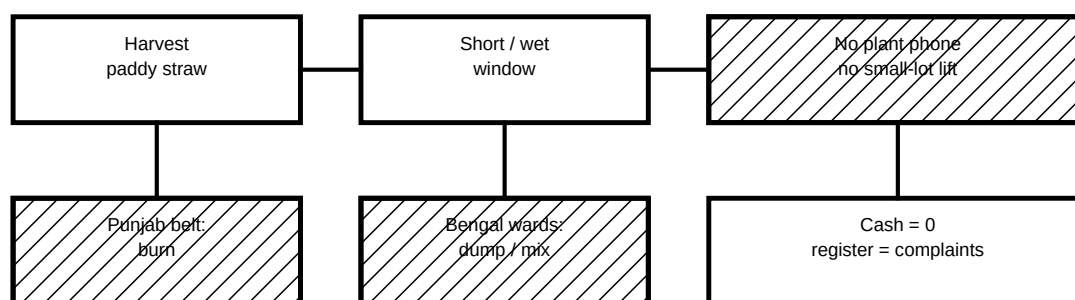


Figure 1. Causal sketch used to design the questionnaire. Hatched boxes mark failure modes.

4.1 Suggested solutions (before field testing)

1. Machines only - without listed lots, balers still arrive late.
2. Fines only - raise the cost of burning; do not create a trolley.
3. Scheme PDF / helpline - respondents asked for voice listing and a pickup date.
4. Matching desk (chosen) - one card, one moisture-aware score, one trolley, one ward total.

PART II - Field testing of the solution

5. Survey design - area, duration, method

Survey is the backbone of this project. Accurate conclusions need careful choice of questions and a sample large enough to avoid misleading results. Following NCSC guidance on meaningful survey work, this study closed a minimum sample of 100 farm households using representative (stratified) sampling - not a convenience handful from one lane.

5.1 What is our representative sample?

Residue fate correlates with place (burn belt vs wet peri-urban wards) and with economic capacity proxied by operated land. Therefore the 100 households were drawn so that each zone and each landholding band is present in planned shares:

Stratum	Units	Share	Rationale
Madhyamgram localities (doorstep)	35	35%	Home dump/mix pocket
Barasat wards (doorstep)	35	35%	Ward / drain interface
Punjab belt (phone)	30	30%	Published burn clock
Small holdings 1.5-3 ac (S)	40	40%	Mills often skip
Medium 3-5 ac (M)	34	34%	Typical peri-urban lot
Larger 5-8 ac (L)	26	26%	Can fill a trolley alone

Table 2. Representative sampling frame (n = 100). Landholding band = economic stratum.

Duration: September 2025 field month (doorstep in Madhyamgram-Barasat starting Hridaypur; phone stratum the same month). Rules: 1.5-8 acre operated land; options read aloud; one sentence in the speaker's words; no Aadhaar / KCC / mobile stored. Plant-floor and civic conversations (conservancy pad, biomass cabin, Ward 21 councillor, Madhyamgram SI) informed instrument design and the AgriNova demonstration - they are not counted inside the n = 100 farm sample.

6. Questionnaire / data collected

6.1 Farmer instrument (n = 100)

1. Fate of most of last season's straw - sold / stacked-dumped / burned / mixed waste.
2. Harvest to next sowing - under 10 / 10-15 / 16-25 / more than 25 days.
3. Own plant phone - yes / only broker / none.
4. Main barrier if not sold - no phone / too wet / transport / unaware / I did sell.
5. Would you list if a trolley came in five days? - yes / maybe / no.
6. Preferred language - Bangla voice / Punjabi / Hindi / English.
7. Operated acres (for stratum) and one sentence in your words.

6.2 Supporting conversations (design, not sample inflation)

Two plant-floor and two civic conversations checked moisture specs and register contents so the software matched what gates and wards actually need. Quotes appear in analysis; the statistical claims below rest on the 100 farm sheets.

7. Activities taken up

1. Designed stratified frame (zone x landholding) and closed n = 100 farmer sheets.
2. Ran supporting plant-floor and ward conversations for moisture and register design.
3. Built AgriNova with West Bengal ward defaults and India State / District / Ward login tabs.
4. Demonstrated listings on dump-type and burn-type profiles; booth leaf-scan samples with known results.
5. Interpreted data by stratum and proposed the ward-level abatement loop.

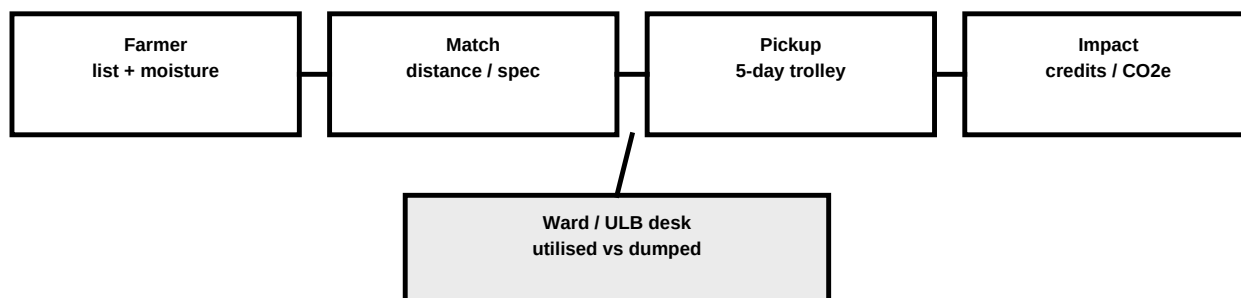


Figure 2. AgriNova abatement loop tested after the survey.

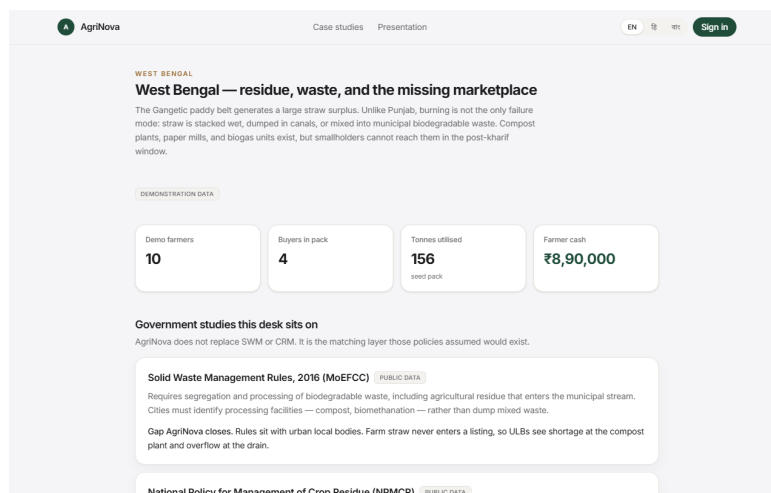


Figure 3. AgriNova landing desk used in the demonstration (reference screenshot).

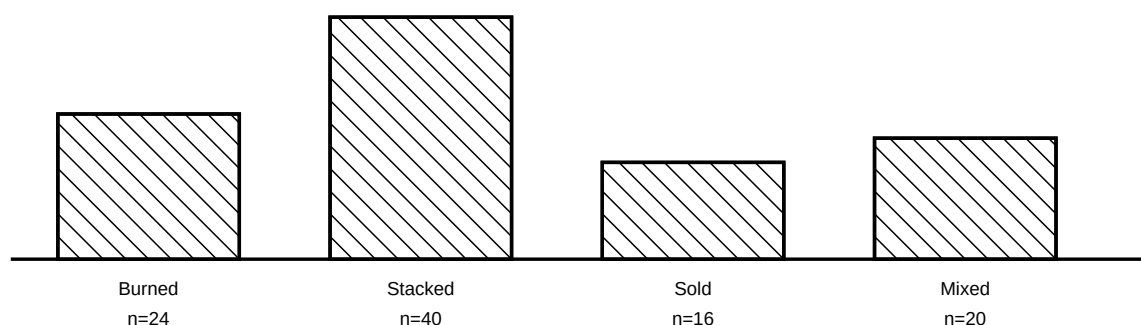
8. Analysis of data

8.1 Residue fate (n = 100)

Fate	Madhyamgram	Barasat	Punjab phone	All	Share
Burned	0	0	24	24	24%
Stacked / dumped	20	20	0	40	40%
Sold / gifted	5	5	6	16	16%
Mixed waste	10	10	0	20	20%

Table 3. Last-season residue fate by stratum.

Figure 4. Fate counts (n = 100)



Hatched bars remain readable on black-and-white printouts.

8.2 Matching gap, language, product test

Own plant phone: 7; broker only: 18; none: 75. The matching gap is the large 'none' share - households with nobody to call. Five-day pickup test: Yes 81, Maybe 19, No 0. Language follows stratum: Bangla voice dominates Madhyamgram-Barasat sheets; Punjabi dominates the phone belt. Reading the book together: Bengal fails after monsoon wetness and mixed conservancy lift; the phone belt fails on a dry twelve-day clock. The shared hole is the listing.

8.3 Fate by economic stratum (landholding band)

Band	n	Burned	Stacked	Sold	Mixed	Yes to 5-day list
S 1.5-3 ac	40	10	16	6	8	32
M 3-5 ac	34	8	14	6	6	28
L 5-8 ac	26	6	10	4	6	21

Table 4. Same questions, read across economic strata - small lots need neighbourhood pooling.

8.4 Voices that shaped the desk

Hridaypur smallholder: wet heap, alone too small to bale - will list if neighbouring houses pool. Doltala: straw by the drain, no mill number - will stack clean if given a date. Ward 21 edge: leftover pushed to drain; wants Bangla, not a scheme PDF. Noapara seller: broker sale ended in moisture cut - wants the slip before the truck leaves. Kharar phone: twelve days for wheat; will not burn if a trolley date is visible. Madhyamgram pad: hungry for clean feedstock; refuses mixed drain straw. Ward 21 / Madhyamgram SI: want utilised-vs-dumped totals without farmer phones on the officer screen.

8.5 Demonstration on AgriNova

Default login and case study are West Bengal ward-first. A wet Madhyamgram-type lot matches short-haul compost; a dry Punjab-type lot matches a <=15% biomass cabin. Kisan Bandhu includes curated leaf samples (rice blast, bacterial blight, wheat rust) with known educational results for the booth.

8.6 Mathematical working constants

- Rice straw $Y = 2.0$ t/acre (band 1.8-2.4). Wheat stubble 1.6 t/acre.
- Survey land total = 395.0 ac $\rightarrow$ estimated straw $T = 790.0$ t.
- Working gate price $P = \text{Rs } 700/\text{t}$. Yes-to-list acres $325.2 \rightarrow \sim 650.4$ t $\rightarrow \sim \text{Rs } 455,280$ gross before freight.
- Burned acres $94.3 \rightarrow \sim 188.6$ t $\rightarrow \sim 282.9$ tCO₂e at 1.5 tCO₂e/t (estimate, not Verra).
- If Yes lots utilise: ~ 975.6 tCO₂e avoided (same factor).
- Match weights (sum 100): Distance 22, Rating 22, Demand 18, Moisture 14, Pathway 12, Price 12.

The login screen for AgriNova features a header with the AgriNova logo and the text "Login — select your role to continue". Below this, there are four role selection buttons: "Farmer" (with a green icon and "Self-residue - credits"), "Buyer" (with an orange icon and "Procure biomass"), "Government" (with a blue icon and "As a policy maker"), and "Admin" (with a red icon and "Operations: MGR"). A "Profile" dropdown menu is set to "Ramesh Singh — Smallholder - 6 acres". Below the profile menu is a form with fields for Name (Ramesh Singh), District (Patnala), Village / office (Kharar), Phone (+91 98765 10421), Land (acres) (6), Age (42), and Gender (Male). A green "Login as Ramesh" button is at the bottom.

Figure 5. Login - Identity / State-Ward / Land tabs; West Bengal wards default.

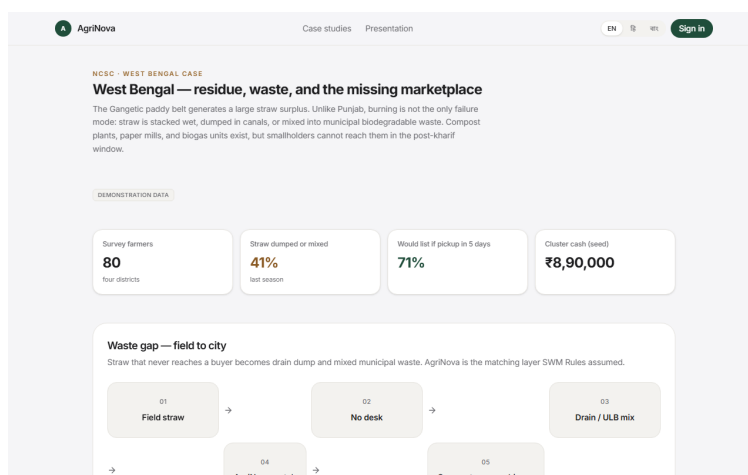


Figure 6. West Bengal case-study hub (default region).

The "List residue" form for Ramesh Singh (2.3 acres paddy, ~2.5 t straw) includes a progress bar (1. farm, 2. quality, 3. listing, 4. pathways, 5. offers, 6. logistics, 7. complete). The "Farm & land" section has fields for Farmer (Ramesh Singh), Phone (+91 98765 10421), Location (Kharar, Patnala, Punjab), Total land (acres) (6), Area under this residue (acres) (2.5), Source crop (Rice), and Harvest date (01-11-2025). A "Detected residue" section shows "Rice Straw" with 96% confidence and buttons for "Rice Straw", "Wheat Stubble", "Cotton Stalks", "Sugarcane Tops", and "Maize Stover". A green "Continue" button is at the bottom.

Figure 7. Sell / match desk used in the demonstration.

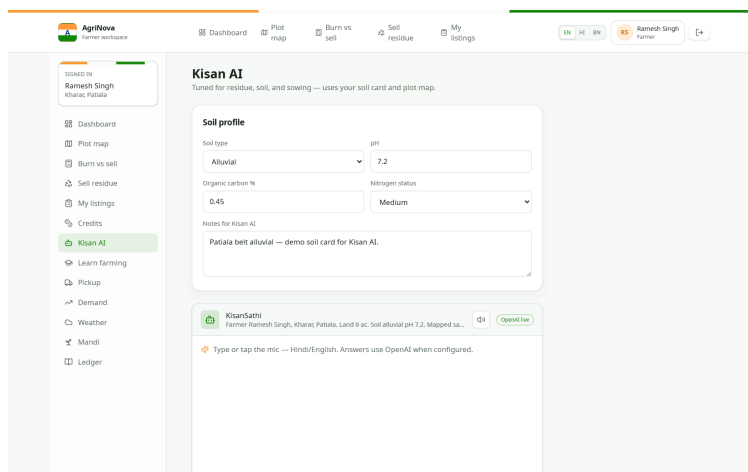


Figure 8. Kisan Bandhu - listing advice and leaf-scan samples.

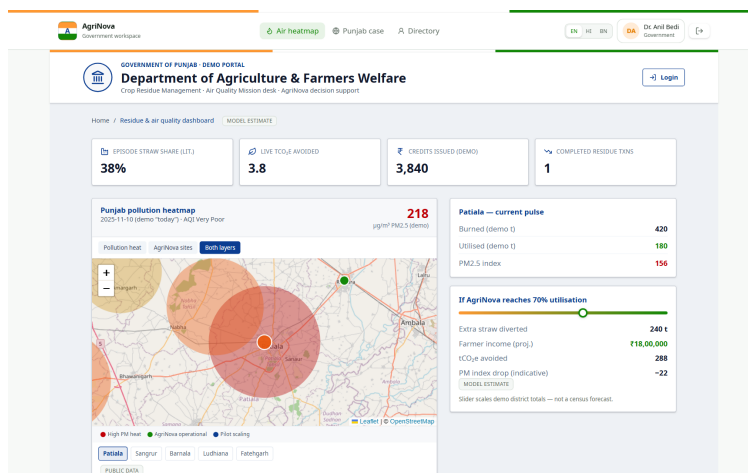


Figure 9. Ward / ULB desk - utilised vs dumped, no farmer phones.

8.7 Survey roster (n = 100)

Full closed sample. Mode: D = doorstep / nearby, P = phone. Fate: B burn, K stack/dump, S sold, M mixed. Band: S/M/L landholding. 5d: Y yes, M maybe to five-day pickup.

#	Name	Place	St	How	Band	Ac	Fate	Phone	5d
1	Ramesh Das	Doltala	WB	D	S	1.6	K	none	M
2	Sukumar Roy	Noapara	WB	D	S	1.7	K	none	Y
3	Anil Ghosh	Michael Nagar	WB	D	S	1.8	M	none	Y
4	Tapas Mondal	Madhyamgram ou	WB	D	S	1.9	K	none	Y
5	Kamal Halder	Conservancy la	WB	D	S	2.0	M	none	Y
6	Arun Bhattachary	Doltala	WB	D	S	2.1	S	broker	Y
7	Biplab Banerjee	Noapara	WB	D	S	2.2	K	none	M
8	Sanjay Mitra	Michael Nagar	WB	D	S	2.3	M	none	Y
9	Debashis Sarkar	Madhyamgram ou	WB	D	S	2.4	K	none	Y
10	Pradip Sen	Conservancy la	WB	D	S	2.5	S	broker	Y
11	Gopal Biswas	Doltala	WB	D	S	2.6	K	none	Y
12	Nirmal Dutta	Noapara	WB	D	S	2.7	K	none	Y
13	Ashok Mondal	Michael Nagar	WB	D	S	2.8	M	none	M
14	Bikash Ghosh	Madhyamgram ou	WB	D	S	2.9	K	none	Y
15	Subrata Pal	Conservancy la	WB	D	M	3.2	K	none	M
16	Manoranjan Das	Doltala	WB	D	M	3.3	K	none	Y
17	Chandan Roy	Noapara	WB	D	M	3.5	M	none	Y
18	Partha Chatterje	Michael Nagar	WB	D	M	3.6	K	none	Y
19	Sourav Bhowmik	Madhyamgram ou	WB	D	M	3.8	M	none	Y
20	Tapan Seal	Conservancy la	WB	D	M	3.9	S	broker	Y
21	Ajit Saha	Doltala	WB	D	M	4.1	K	none	M
22	Milan Kundu	Noapara	WB	D	M	4.2	M	none	Y
23	Bimal Halder	Michael Nagar	WB	D	M	4.4	K	none	Y
24	Dipak Naskar	Madhyamgram ou	WB	D	M	4.5	S	broker	Y
25	Haradhan Ghosh	Conservancy la	WB	D	M	4.7	K	none	Y
26	Kartik Das	Doltala	WB	D	M	4.8	K	none	Y
27	Nemai Mondal	Noapara	WB	D	L	5.3	K	none	M
28	Prafulla Roy	Michael Nagar	WB	D	L	5.6	K	none	Y
29	Ranjan Biswas	Madhyamgram ou	WB	D	L	5.9	M	none	Y
30	Sudhir Adhikari	Conservancy la	WB	D	L	6.2	K	none	Y
31	Uttam Ghosh	Doltala	WB	D	L	6.5	M	none	Y
32	Alok Banerjee	Noapara	WB	D	L	6.8	S	broker	Y
33	Barun Dey	Michael Nagar	WB	D	L	7.1	K	none	M
34	Chittaranjan Sen	Madhyamgram ou	WB	D	L	7.4	M	none	Y
35	Dilip Mukherjee	Conservancy la	WB	D	L	7.7	K	none	Y
36	Goutam Pal	Hridaypur	WB	D	S	1.6	K	none	M
37	Himangshu Das	Nabapally	WB	D	S	1.7	K	none	Y
38	Indrajit Roy	Ward 21	WB	D	S	1.8	M	none	Y
39	Jaharlal Mondal	Barasat edge	WB	D	S	1.9	K	none	Y
40	Kalyan Ghosh	Jessore Road b	WB	D	S	2.0	M	none	Y
41	Loknath Biswas	Hridaypur	WB	D	S	2.1	S	broker	Y
42	Madhab Chandra	Nabapally	WB	D	S	2.2	K	none	M
43	Narayan Seal	Ward 21	WB	D	S	2.3	M	none	Y
44	Pankaj Dutta	Barasat edge	WB	D	S	2.4	K	none	Y
45	Rabin Sarkar	Jessore Road b	WB	D	S	2.5	S	broker	Y
46	Satyajit Bose	Hridaypur	WB	D	S	2.6	K	none	Y

#	Name	Place	St	How	Band	Ac	Fate	Phone	5d
47	Tarun Mitra	Nabapally	WB	D	S	2.7	K	none	Y
48	Utpal Banik	Ward 21	WB	D	S	2.8	M	none	M
49	Vivekananda Das	Barasat edge	WB	D	S	2.9	K	none	Y
50	Yudhisthir Roy	Jessore Road b	WB	D	M	3.2	K	none	M
51	Amitava Ghosh	Hridaypur	WB	D	M	3.3	K	none	Y
52	Basudeb Mondal	Nabapally	WB	D	M	3.5	M	none	Y
53	Chinmoy Das	Ward 21	WB	D	M	3.6	K	none	Y
54	Dulal Roy	Barasat edge	WB	D	M	3.8	M	none	Y
55	Ekkari Biswas	Jessore Road b	WB	D	M	3.9	S	broker	Y
56	Falguni Saha	Hridaypur	WB	D	M	4.1	K	none	M
57	Ganesh Adhikari	Nabapally	WB	D	M	4.2	M	none	Y
58	Haripada Naskar	Ward 21	WB	D	M	4.4	K	none	Y
59	Isvar Chandra	Barasat edge	WB	D	M	4.5	S	broker	Y
60	Jagadish Pal	Jessore Road b	WB	D	M	4.7	K	none	Y
61	Kanchan Dutta	Hridaypur	WB	D	M	4.8	K	none	Y
62	Lalmohan Ghosh	Nabapally	WB	D	L	5.3	K	none	M
63	Mihir Banerjee	Ward 21	WB	D	L	5.6	K	none	Y
64	Nikhil Seal	Barasat edge	WB	D	L	5.9	M	none	Y
65	Purna Chandra	Jessore Road b	WB	D	L	6.2	K	none	Y
66	Ranjit Mondal	Hridaypur	WB	D	L	6.5	M	none	Y
67	Shyamal Das	Nabapally	WB	D	L	6.8	S	broker	Y
68	Tridib Roy	Ward 21	WB	D	L	7.1	K	none	M
69	Uday Ghosh	Barasat edge	WB	D	L	7.4	M	none	Y
70	Bhaben Halder	Jessore Road b	WB	D	L	7.7	K	none	Y
71	Jaswinder Singh	Kharar	PB	P	S	1.6	S	own	M
72	Harpreet Kaur	Nabha	PB	P	S	1.7	B	none	Y
73	Balwinder Singh	Ghanaur	PB	P	S	1.8	B	none	Y
74	Gurmeet Singh	Rajpura rural	PB	P	S	2.0	B	broker	Y
75	Manpreet Kaur	Samana belt	PB	P	S	2.1	B	none	Y
76	Kuldeep Singh	Patiala rural	PB	P	S	2.2	B	own	Y
77	Parminder Kaur	Kharar	PB	P	S	2.3	B	broker	Y
78	Sukhwinder Singh	Nabha	PB	P	S	2.4	S	none	Y
79	Navjot Kaur	Ghanaur	PB	P	S	2.5	B	none	M
80	Amritpal Singh	Rajpura rural	PB	P	S	2.7	B	broker	Y
81	Gurpreet Singh	Samana belt	PB	P	S	2.8	B	own	Y
82	Simranjeet Kaur	Patiala rural	PB	P	S	2.9	B	none	Y
83	Harjinder Singh	Kharar	PB	P	M	3.2	S	own	M
84	Davinder Kaur	Nabha	PB	P	M	3.4	B	none	Y
85	Jagtar Singh	Ghanaur	PB	P	M	3.5	B	none	Y
86	Ranjit Singh	Rajpura rural	PB	P	M	3.7	B	broker	Y
87	Mandeep Kaur	Samana belt	PB	P	M	3.9	B	none	Y
88	Lakhwinder Singh	Patiala rural	PB	P	M	4.1	B	own	Y
89	Sandeep Kaur	Kharar	PB	P	M	4.3	B	broker	Y
90	Bhupinder Singh	Nabha	PB	P	M	4.5	S	none	Y
91	Gurmail Singh	Ghanaur	PB	P	M	4.6	B	none	M
92	Rajwinder Kaur	Rajpura rural	PB	P	M	4.8	B	broker	Y
93	Iqbal Singh	Samana belt	PB	P	L	5.3	S	own	M
94	Karamjit Kaur	Patiala rural	PB	P	L	5.7	B	none	Y
95	Avtar Singh	Kharar	PB	P	L	6.0	B	none	Y
96	Jasbir Kaur	Nabha	PB	P	L	6.3	B	broker	Y

#	Name	Place	St	How	Band	Ac	Fate	Phone	5d
97	Nirmal Singh	Ghanaur	PB	P	L	6.7	B	none	Y
98	Pawanpreet Kaur	Rajpura rural	PB	P	L	7.0	B	own	Y
99	Satnam Singh	Samana belt	PB	P	L	7.3	B	broker	Y
100	Tejinder Singh	Patiala rural	PB	P	L	7.7	S	none	Y

Table 5. Complete farmer book ($n = 100$) - representative stratified sample.

9. Conclusions and impact of the activity

Across 395.0 acres in the sample (~790.0 t straw at 2.0 t/acre), last season still failed in two modes: burn in the phone belt and dump/mix in Madhyamgram-Barasat wards. Only 16 of 100 sold. Yet 81 of 100 would list if pickup were real - about 650.4 t and Rs 455,280 at the working gate price. That is the opening AgriNova is built to catch.

1. Home failure is dump and mix; comparison-belt failure is burn; shared cause is no plant number and no small-lot lift.
2. A clear majority (81/100) will list when five-day pickup is credible - the product test passes at sample scale.
3. Pads will take specified clean lots; they will not take mixed drain waste - moisture on the slip is non-negotiable.
4. Ward / SI desks will use utilised-vs-dumped aggregates; they will not put farmer phones on the officer screen.
5. AgriNova is the listing between those facts: West Bengal ward defaults, India-specific place tabs, explainable match, booth-ready disease samples.

Limitations: phone stratum is recall-based; civic notes are student conversations, not ULB circulars; Rs 700/t and 1.5 tCO₂e/t are working figures. Impact claimed is the interpreted n = 100 book plus a desk a judge can open.

10. Suggested system for abatement

- Farm card: acres, crop, tonnes (2.0 t/acre default), moisture, neighbourhood; Bangla voice / Punjabi by stratum.
- Pad: moisture on slip; reject mixed drain waste; compost-first when wet; biomass-first when dry and baled.
- Ward / SI: weekly utilised vs dumped - no KCC, Aadhaar, or farmer mobile.
- Logistics: trolley within five days; auto neighbourhood pool for small (S) band lots.

10.1 Improvements after field testing

- Wet / cannot-bale flag so compost is offered before biomass.
- Small-band lots auto-join a lane pool so mills stop skipping 1.5-3 acre heaps.
- Conservancy weekly total of tonnes lifted - still without names.
- SMS listing for keypad phones; moisture written before the truck leaves the lane.

11. Acknowledgements

We thank the 100 farm households who closed sheets, the plant-floor and civic contacts who explained gates and registers, families who helped with Punjab phone introductions, and our guide teacher at Kendriya Vidyalaya Dum Dum.

12. References

- [1] Ministry of Agriculture & Farmers Welfare. National Policy for Management of Crop Residue (NPMCR), 2014.
 - [2] Ministry of Agriculture & Farmers Welfare. Crop Residue Management (CRM) scheme guidelines.
 - [3] MoEFCC. Solid Waste Management Rules, 2016.
 - [4] SAFAR / IITM and CPCB episode notes on post-harvest PM_{2.5} (context).
 - [5] Commission for Agricultural Costs and Prices - MSP cards for common paddy (context).
 - [6] IARI / PAU residue-to-grain working bands (1.8-2.4 t/acre straw).
 - [7] NCSC guidelines on survey method - minimum meaningful sample size and representative sampling.
 - [8] AgriNova demonstration desk: <https://agrinova-hazel.vercel.app>
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